

Hugging Face & On-Device Models Cheat Sheet

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Baseline (checked 2026-09-30): transformers 5.17.0 · huggingface_hub 2.0.0 (hf CLI) · peft 0.21.1 · trl 1.14.1 · datasets 5.0.1 · bitsandbytes 0.50.2 · sentence-transformers 6.1.0 · Gradio 6 · smolagents 1.26.0 · optimum-onnx 0.1.0 · ONNX Runtime 1.30.0 · llama-cpp-python 0.3.35 · mlx-lm 0.31.3 · faster-whisper 1.2.1 · Transformers.js 4.3.0 · LiteRT 2.2.0 · LiteRT-LM 0.17.1 · swift-transformers 1.3.4 · mlx-swift-lm 3.31.4 · argmax-oss-swift 1.1.0 · Python 3.12 · Java 21 · Spring Boot 4.1 · Spring AI 2.0.1. Token always \$HF_TOKEN, never in code. No benchmark numbers or prices: measure on your data, re-read the model card (licences and gating change). Dated 2026-09-30.

HF CLI (WAS HUGGINGFACE-CLI)

```
hf auth login [--token "$HF_TOKEN"]
hf auth whoami | logout
hf download <repo> [files] --include "*"*.safetensors"
--exclude _ --repo-type dataset --revision main
--local-dir ./m --cache-dir _ --dry-run
hf models ls --pipeline-tag automatic-speech-recognition
--no-gated --sort downloads --limit 10 [--json]
hf models info | card <id> --metadata -
--format json
hf cache ls [--filter "size>1g"] | prune | rm | verify
```

pip install -U huggingface_hub. Gated: accept terms in the browser, download with that account's token. A gate is not a licence.

DOWNLOAD API · ENV VARS

```
from huggingface_hub import
(hf_hub_download,
 snapshot_download)
p = hf_hub_download(repo_id="ggml-
org/gemma-3-1b-it-GGUF",
 filename="gemma-3-1b-it-
Q4_K_M.gguf") # 1 file
d =
snapshot_download(repo_id="openai/whis
per-small.en",
 allow_patterns=
["*.safetensors","*.json"])
```

Also revision, local_dir, token, dry_run=True. Cache files are read-only.

Variable	Meaning
HF_HOME	root, default ~/.cache/huggingface
HF_HUB_CACHE	repo cache, \$HF_HOME/hub
HF_TOKEN	token; beats hf auth login
HF_HUB_OFFLINE=1	no Hub calls (local_files_only=True)
HF_XET_HIGH_PERFORM ANCE=1	faster Xet transfers
HF_HUB_DISABLE_XET= 1	non-Xet fallback
HF_HUB_DISABLE_SYML INKS=1	no symlinks

Set before importing huggingface_hub. Offline: download online first, then HF_HUB_OFFLINE=1.

HOSTED FALLBACK

```
OpenAI(base_url="https://router.hugging
face.co/v1",
 api_key=os.environ["HF_TOKEN"])
model="Qwen/Qwen3-8B:cheapest"
```

Suffixes :fastest :cheapest :preferred :<provider>; check the model's Providers panel.

LICENCE CHECKLIST

- Repo + commit hash, pinned via revision=
- Weights licence: read the LICENSE file
- Commercial use (revenue, user count)?
- Redistribution: bundle or host; derivatives?
- Attribution and notices
- Use restrictions (AUP, OpenRAIL)
- Copyleft reach (AGPL pulls in your app)
- Gating: which account; no token in an app
- Training-data and output terms
- Conversions: base_model licence of every GGUF / MLX / ONNX / LiteRT variant
- Archive licence text and card at release

Classes: permissive (Apache-2.0, MIT) · attribution (CC-BY-4.0) · OpenRAIL · community (Llama, Gemma) · **AGPL-3.0 · non-commercial**. Quants inherit the base licence.

FORMAT → RUNTIME

Format	Read by (Hub org)
.safetensors	transformers, diffusers, vLLM, MLX
.gguf	llama.cpp, Ollama, LM Studio (ggml-org)
.onnx	ONNX Runtime, Transformers.js (onnx-community)
MLX	mlx-lm, mlx-swift-lm (mlx-community)
.mlpackage	Core ML: CPU, GPU, Neural Engine (apple)
.aimodel	Core AI, iOS / macOS 27+ (convert yourself)
.tflite	LiteRT (litert-community)
.litertlm	LiteRT-LM, on-device LLMs (litert-community)
.pte	ExecuTorch (pytorch, software-mansion)
model.bin	CTranslate2, faster-whisper (Systran)

Pickle can run code on load: prefer safetensors, torch.load(..., weights_only=True); trust_remote_code only for code you read, pinned. iPhone → Core ML / MLX · Android → LiteRT / ONNX · browser → ONNX · Mac LLM → GGUF · Python server → safetensors / CT2 · JVM, .NET → ONNX Runtime.

PIPELINE TASKS (V5)

```
asr = pipeline(task="automatic-speech-
recognition",
 model="openai/whisper-large-v3-
turbo",
 dtype=torch.float16)
```

automatic-speech-recognition · text-to-audio (alias text-to-speech) · image-classification · zero-shot-image-classification · object-detection · image-segmentation · mask-generation · image-text-to-text · text-generation. Always pass model=; device or device_map="auto", not both. **Removed:** summarization, translation, question-answering, image-to-text, visual-question-answering, image-to-image → chat model or Auto*. Local OpenAI API: transformers serve Qwen/Qwen3-0.6B (transformers[serving], localhost:8000, no auth).

CONVERSION ONE-LINERS

```
GGUF python convert_hf_to_gguf.py ./m
--outfile m.gguf --outtype bf16
./llama-quantize m.gguf m-q4.gguf
Q4_K_M
ONNX optimum-cli export onnx --model
<id>
--dtype fp16 out/ # optimum-
onnx[onnxruntime]
python -m
onnxruntime_genai.models.builder
-m <id> -o out -p int4 -e cpu
MLX mlx_lm.convert --hf-path <id> -q --
q-bits 4
LiteRT-LM litert-torch export_hf --
model=<id>
--output_dir=out
ExecuTorch optimum-cli export
executorch
--model <id> --recipe xnnpack --
qlinear 8da4w
CT2 ct2-transformers-converter --model
<id>
--output_dir o --quantization float16
Core ML ct.convert(tored, convert_to=
"mlprogram") → coremltools.optimize
Core AI torch.export → coreai-torch →
.aimodel
GPU bitsandbytes NF4 · GPTQModel · llm-
compressor
Windows winml export | optimize |
quantize | compile
```

A conversion is a derivative: same licence. Compare with the original; keep tokenizer and chat template.

LORA · QLORA · TRL SFT

$W' = W + \frac{\alpha}{r} B A, B \in \mathbb{R}^{d_{out} \times r}, A \in \mathbb{R}^{r \times d_{in}}$

B starts at 0, so $W' = W$ at step 0. Trainable: $r(d_{in}+d_{out})$ vs $d_{in}d_{out}$; use_rslora=True: a/r . QLoRA = NF4 base + LoRA (CUDA).

```
args = SFTConfig(output_dir="out",
 learning_rate=1e-4, max_length=2048,
 packing=True,
 assistant_only_loss=True)
tr = SFTTrainer(model="Qwen/Qwen3-
0.6B",
 args=args, train_dataset=ds,
 peft_config=LoraConfig(r=16,
 lora_alpha=32,
 target_modules="all-linear",
 task_type="CAUSAL_LM"))
tr.train(); tr.save_model() #
adapter only
m = PeftModel.from_pretrained(base,
"out")
m.merge_and_unload().save_pretrained("m
erged")
python convert_lora_to_gguf.py ./out --
base ./m
```

Ladder: prompt → retrieve → fine-tune. Rank 8–64. Merge into a bf16 base, never quantized weights; merged model keeps the base licence. Re-evaluate after each quantize or export.

WEB · JS

```
import { pipeline } from
"@huggingface/transformers";
const asr = await pipeline(
"automatic-speech-recognition",
"onnx-community/whisper-tiny.en",
{ device: gpu ? "webgpu" : "wasm",
 dtype: gpu ? "q4f16" : "q8" });
```

onnxruntime-web; React Native: react-native-executortorch, llama.m.

DEPRECATIONS & RENAMES

Old	Now
huggingface-cli	hf
api- inference.huggingface.co (decommissioned)	Inference Providers
HUGGINGFACE_HUB_CACHE , HUGGING_FACE_HUB_TOKE N, TRANSFORMERS_CACHE	HF_HUB_CACHE, HF_TOKEN, HF_HOME
HF_HF_ENABLE_HF_TRAN SFER	Xet
torch_dtype, use_auth_token	dtype, token
load_in_4bit=on from_pretrained	BitsAndBytesConfig
TF / Flax; AutoModelWithLMHead	PyTorch only; AutoModelForCausalL M
TRL tokenizer=, max_seq_length	processing_class, max_length
ONNX export in optimum	optimum-onnx
TensorFlow Lite, ai- edge-torch	LiteRT, litert-torch
MediaPipe LLM Inference (maintenance).task	LiteRT-LM, .litertlm
WhisperKit repo; mlx- swift-examples LLM libs	argmax-oss-swift; mlx-swift-lm
AutoAWQ (deprecated)	llm-compressor
DirectML (sustained engineering); NNAPI (Android 15)	Windows ML; LiteRT
Phi Silica	Aion Instruct per vendor banner; re- check dates

PYTHON · FASTAPI

```
@asynccontextmanager
async def lifespan(app): # once per
worker
    app.state.asr = WhisperModel("turbo",
 device="cpu", compute_type="int8")
    app.state.llm =
Llama.from_pretrained(
    repo_id="Qwen/Qwen3-1.7B-GGUF",
    filename="*Q8_0.gguf", n_ctx=4096)
    yield
    app = FastAPI(lifespan=lifespan)
```

Lock the Llama (one generation at a time);
run_in_threadpool; ONNX providers=[_,
"CPUExecutionProvider"].

JAVA · SPRING BOOT

```
# spring-ai-starter-model-transformers
spring.ai.model.embedding=transformers
spring.ai.embedding.transformer.onnx.mo
del-url=
classpath:/onnx/all-MiniLM-L6-
v2/model.onnx
spring.ai.embedding.transformer.tokeniz
er.url=
classpath:/onnx/all-MiniLM-L6-
v2/tokenizer.json
spring.ai.model.chat=ollama # LLM:
local server
spring.ai.ollama.chat.model=qwen3:4b
```

Or openai starter + vLLM base-url. In-process:
DJL Criteria, OrtSession.

APPLE · SWIFT

```
let m = try await
#huggingfaceLoadModelContainer(
 configuration:
LLMRegistry.gemma3_1b_qat_4bit)
let a = try await
ChatSession(m).respond(to: q)
let w = try await
WhisperKit(WhisperKitConfig(
 model: "large-v3-v20240930_626MB"))
```

mlx-swift-lm, argmax-oss-swift, swift-
transformers; Core ML, or Core AI (AIModel,
27+).

ANDROID · KOTLIN

```
implementation("com.google.ai.edge.lite
rtlm:
    litertlm-android:0.17.1")
val e = Engine(EngineConfig(
    modelPath = f.absolutePath, backend =
Backend.GPU()))
e.initialize() // not on the
main thread
e.createConversation().sendMessageAsync
(p) // Flow
```

litert-community/gemma-4-E2B-it-litert-
lm; images: LiteRT CompiledModel; Gemini Nano
(beta): keep a fallback.

WINDOWS · C#

```
await
ExecutionProviderCatalog.GetDefault()
.EnsureAndRegisterCertifiedAsync();
opts.SetEpSelectionPolicy(
ExecutionProviderDevicePolicy.MAX_EFFIC
IENCY);
using var s = new
InferenceSession("m.onnx", opts);
IChatClient c = new OllamaApiClient(
    new Uri("http://localhost:11434/"),
    "phi3:mini");
```

ORT GenAI (0.x); Foundry Local: foundry run,
port from foundry server status.

CURATED MODELS (CARD LICENCE, 2026-09-30; QUANTS INHERIT THE BASE LICENCE)			
Type	Model (licence)		Watch out
STT	openai/whisper-large-v3 Apache-2.0 · openai/whisper-large-v3-turbo MIT · openai/whisper-tiny.en Apache-2.0 · Systran/faster-whisper-small MIT · distil-whisper/distil-large-v3.5 MIT · moonshine-ai/moonshine-base MIT · nvidia/parakeet-tdt-0.6b-v3 CC-BY-4.0		CC-BY: credit
TTS	hexgrad/Kokoro-82M Apache-2.0 · suno/bark-small MIT · parler-tts/parler-tts-mini-v1.1 Apache-2.0 · rhasspy/piper-voices MIT repo, per-voice card; engine GPT-3.0 · coqui/XTTS-v2 CPML · facebook/mms-tts-eng CC-BY-NC-4.0		non-commercial
Vision	google/siglip2-base-patch16-224 · facebook/dinov2-base · PekingU/rt detr_v2_r18vd · facebook/sam2.1-hiera-small · HuggingFaceTB/SmolVLM-256M-Instruct · Qwen/Qwen3-VL-2B-Instruct all Apache-2.0 · florence-community/Florence-2-base MIT · apple/FastVLM-0.5B apple-amrl · facebook/sam3 other, gated · Ultralytics/YOLO11 AGPL-3.0		AGPL reaches your app
Small LLM	google/gemma-4-E2B-it · Qwen/Qwen3-0.6B · Qwen/Qwen3.5-2B · HuggingFaceTB/SmolLM3-3B all Apache-2.0 · microsoft/Phi-4-mini-instruct MIT · google/gemma-3-1b-it Gemma terms, gated · meta-llama/Llama-3.2-1B-Instruct Llama 3.2 community, gated		gated: no token in an app
Embeddings	sentence-transformers/all-MiniLM-L6-v2 Apache-2.0 (384 dim) · Qwen/Qwen3-Embedding-0.6B Apache-2.0 · google/embeddinggemma-300m Gemma terms, gated		new model: re-embed